

Factors, Multiples and Patterns — Diagnosing a Rule That Was Guessed From Too Few Terms
Grades 4–5

A rule that fits the first two or three numbers is only a **guess**. A guess that is then used everywhere is an **overgeneralized rule**.

- In each problem, someone states a rule and the evidence they checked.
- Test the rule on a term they did *not* check, or find one **counterexample** — a single case where the rule fails.
- Then answer the question with the *repaired* rule. Show your testing.

1. Pattern A: 3, 6, 9, 12, ...

Kip checks only the first two terms and writes the rule “*double the term before*” — and 3 doubled really is 6.

Test Kip’s rule on the third term. Does doubling 6 give the third term of Pattern A? Then use the pattern’s real rule to find the 7th term.

Answer: _____

2. Pattern B: 1, 4, 9, 16, ...

Dee writes the rule “*add 3*” because $1 + 3 = 4$.

Show that Dee’s rule fails at the third term. The real pattern is the square numbers (1×1 , 2×2 , 3×3 , ...). Find the 7th term.

Answer: _____

3. Find and fix. Nia claims: “*Every multiple of 4 is also a multiple of 8.*” Her evidence is 8, 16 and 24 — all multiples of 8.

Ravi answers that 12 is a counterexample. Divide 12 by 8 and write the **remainder**. A remainder that is not zero proves Nia’s rule is overgeneralized.

Answer: _____

4. Find and fix. Pattern C: 5, 10, 15, 20, ...

Tam writes: “*The terms are the multiples of 5, so every term ends in 5.*” The first half of Tam’s rule is right; the ending-in-5 part is overgeneralized from the first term only.

Cross out the wrong half of Tam’s rule. Then use the correct half to find the 12th term.

Answer: _____

5. Pattern D starts at 4 and adds 4 each time. Bea says the rule is “*double the term before*” — which gives the same first two terms.

Work out the 6th term of Pattern D, and the 6th term Bea’s doubling rule would give. Then write $<$, $>$ or $=$ between them:

Pattern D’s 6th term _____ Bea’s 6th term

6. Find and fix. Ollie claims: “*Every odd number is prime.*” His evidence is 3, 5 and 7. Find the smallest odd number greater than 7 that is *not* prime, and prove it is not prime by writing it as a product of two smaller whole numbers, both greater than 1. Write the product’s value on the answer line.

Answer: _____

7. Pattern E: 1, 2, 4, 7, 11, ...

Ines writes the rule “*double the term before*” because 1 doubles to 2 and 2 doubles to 4. Show the exact term where Ines’s rule first breaks. The real pattern adds 1, then 2, then 3, and so on — one more each step. Find the 8th term.

Answer: _____

8. Explain. Rosa claims: “*Every number that ends in 0 is a multiple of 20.*” Her evidence is 20, 40 and 60.

- (a) Explain, in your own words, why three examples that work do not prove Rosa’s rule.
- (b) Give one counterexample to Rosa’s rule and show why it is a counterexample.
- (c) Write one sentence a classmate could use as a habit for testing any new pattern rule.